GLP-1 scientific abstracts

Project 2 · Experimental report

Corpus: 1,000 English PubMed abstracts. Query: ("GLP-1"[Title/Abstract] OR "glucagon-like peptide-1"[Title/Abstract] OR "glucagon like peptide 1"[Title/Abstract]) AND hasabstract[text] AND english[lang]. The saved selection follows PubMed's returned order, excluding previously archived papers; it is not a random sample. Each PMID is unique and every record has a nonempty English abstract. Only AbstractText content is analyzed, without titles, keywords, section labels or article bodies. Four cumulative conditions use exactly the same raw abstracts. A splits on whitespace and lowercases, keeping attached punctuation. B strips non-alphanumeric boundaries, preserves numeric trailing percent signs and internal punctuation, and normalizes Unicode hyphens/apostrophes. C removes the recorded NLTK English stopword list. D applies the local original Porter 1980 algorithm to ASCII alphabetic tokens. All vocabulary terms are ranked by decreasing CF, with alphabetic tie-breaking. Regression uses unweighted OLS of log10(CF) against log10(rank). Rank regions were fixed before interpreting the fits: first 10%, next 70%, last 20%, with ceiling boundaries. Textbook IDF uses natural logarithms. The corpus snapshot, settings, stopword list, source hashes, exported tables and run metadata support offline reproduction.

Corpus fingerprint

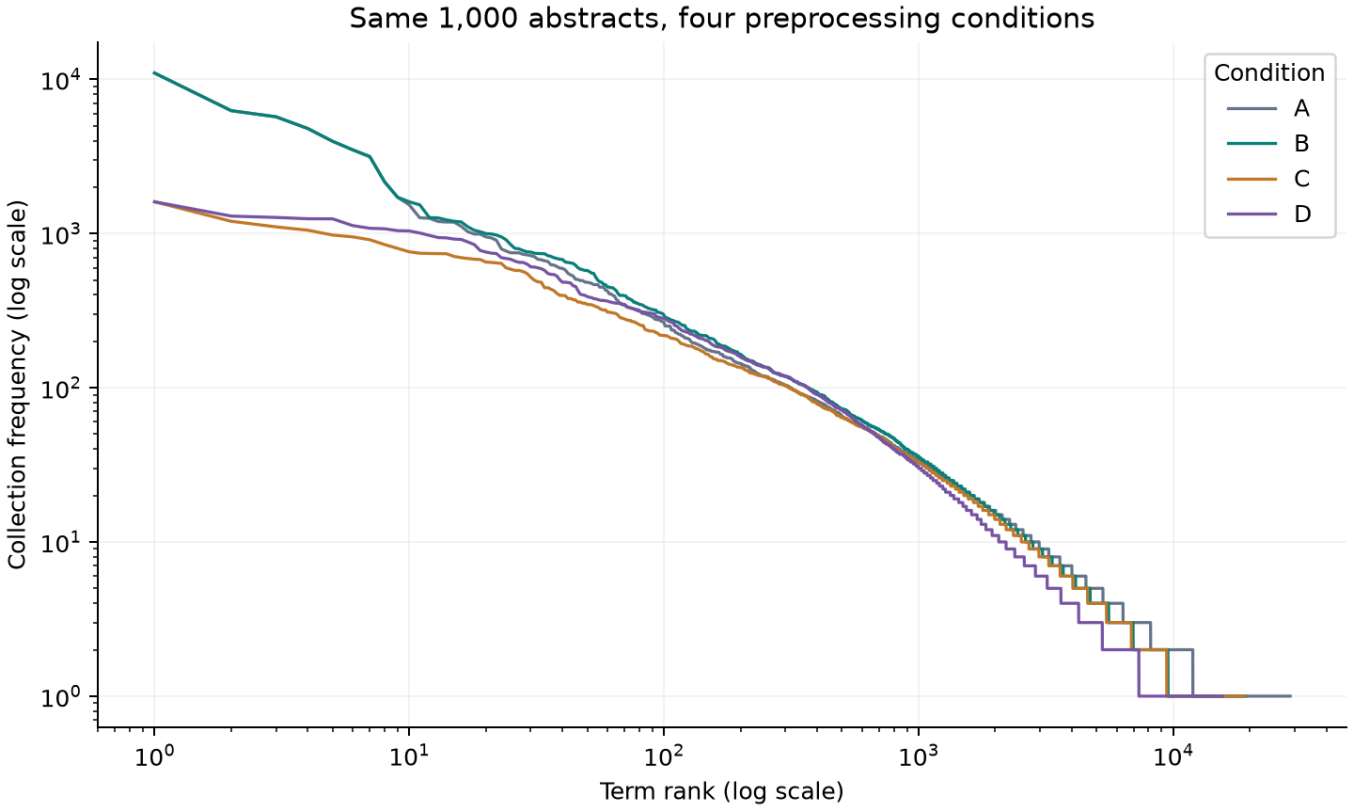
43be97743f89d09e7e8821d116d4a352d3b9b9527196180be215b8a420dd1de1

Evidence package

Tables and plots use one frozen corpus. All numeric claims refer to the recorded condition. The textbook IDF analysis is separate from BM25 retrieval.

Preprocessing and vocabulary

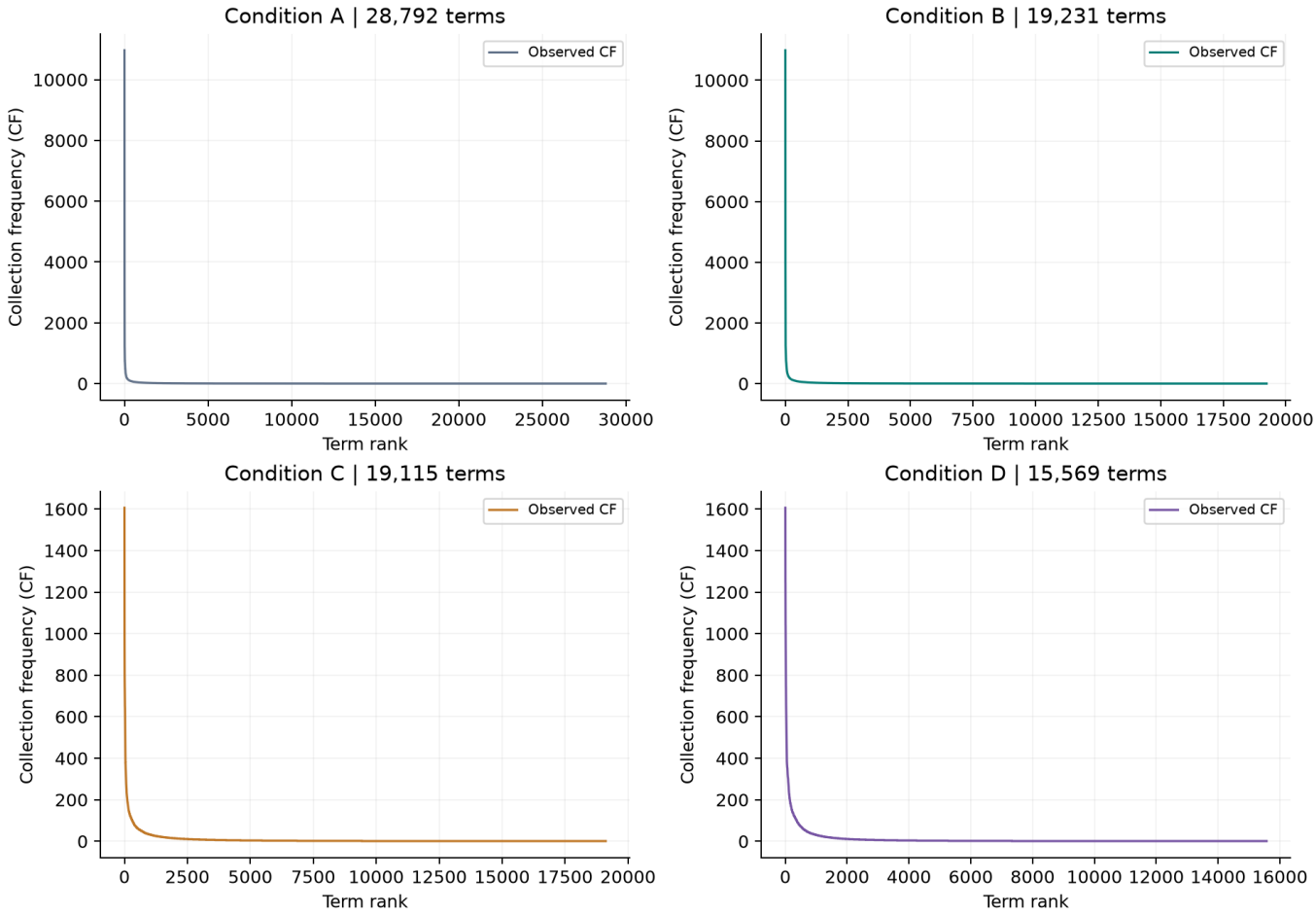
Stage	Tokens	Vocabulary	Mean tokens/doc	Hapax
A	245,416	28,792	245.416	16,855
B	243,843	19,231	243.843	9,692
C	177,052	19,115	177.052	9,688
D	177,052	15,569	177.052	8,241



RQ3 — Preprocessing effects

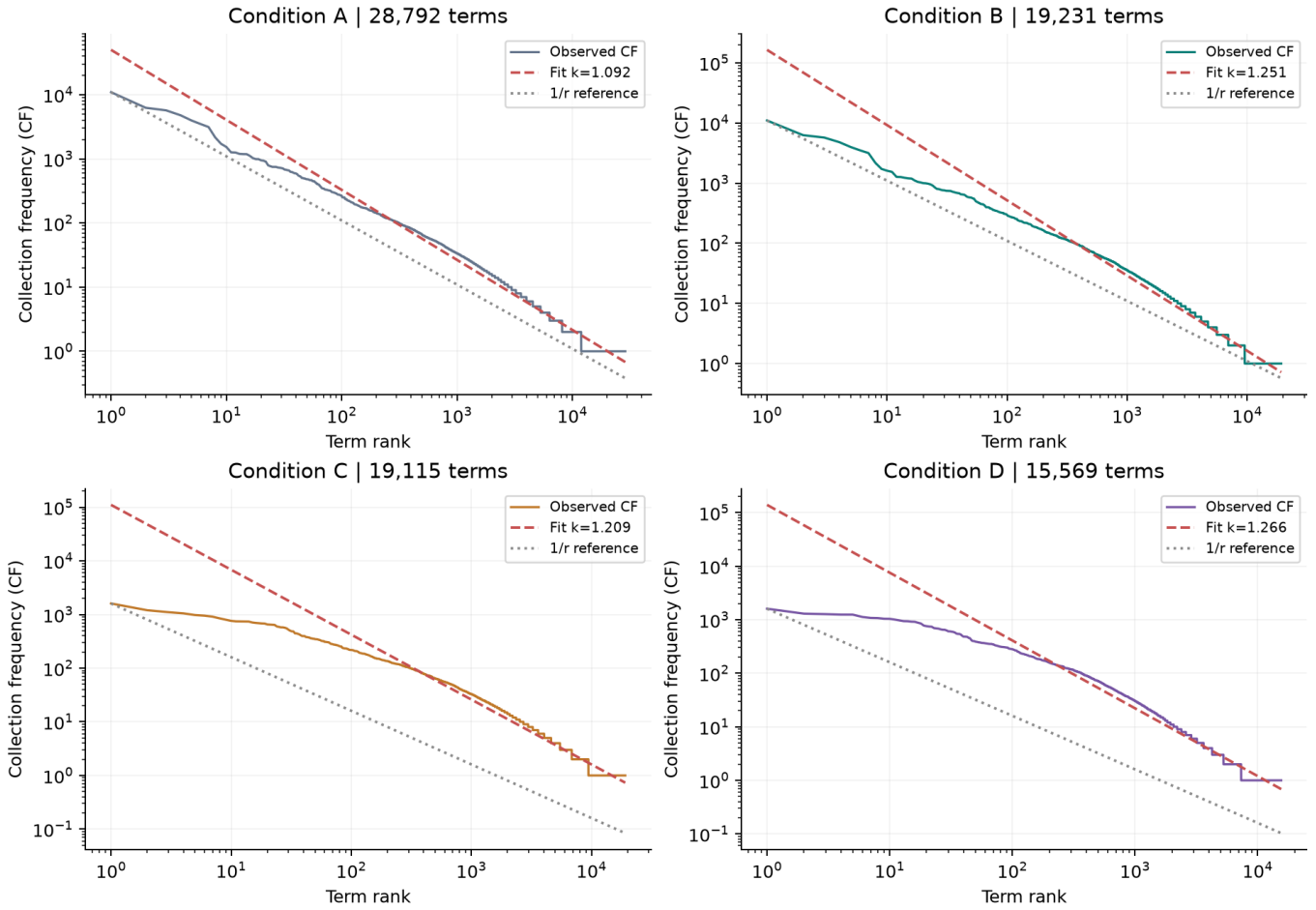
A contains 245,416 tokens and 28,792 distinct terms. Boundary punctuation normalization reduces vocabulary to 19,231 in B, while retaining internal punctuation in biomedical identifiers such as GLP-1. Stopword removal reduces tokens from 243,843 to 177,052; vocabulary becomes 19,115. Porter then reduces vocabulary to 15,569 (18.55% below C), with 177,052 retained tokens. Punctuation merges surface variants, stopwords reshape the head, and stemming combines morphological forms throughout the curve. Neither a smaller vocabulary nor an exponent closer to one proves better retrieval. Inspect the Top 50 tables and merged forms for specific examples.

Rank-frequency distributions



Frequency is CF, summed over the collection. Ranks start at one; alphabetic order breaks ties. The linear view makes the dominance of the head visible.

Log-log regression



Stage	Slope	Intercept	k	R^2	RMSE log10
A	-1.0915	4.6997	1.0915	0.9571	0.1003
B	-1.2508	5.2149	1.2508	0.9687	0.0975
C	-1.2095	5.0433	1.2095	0.9633	0.1024
D	-1.2663	5.1450	1.2663	0.9613	0.1101

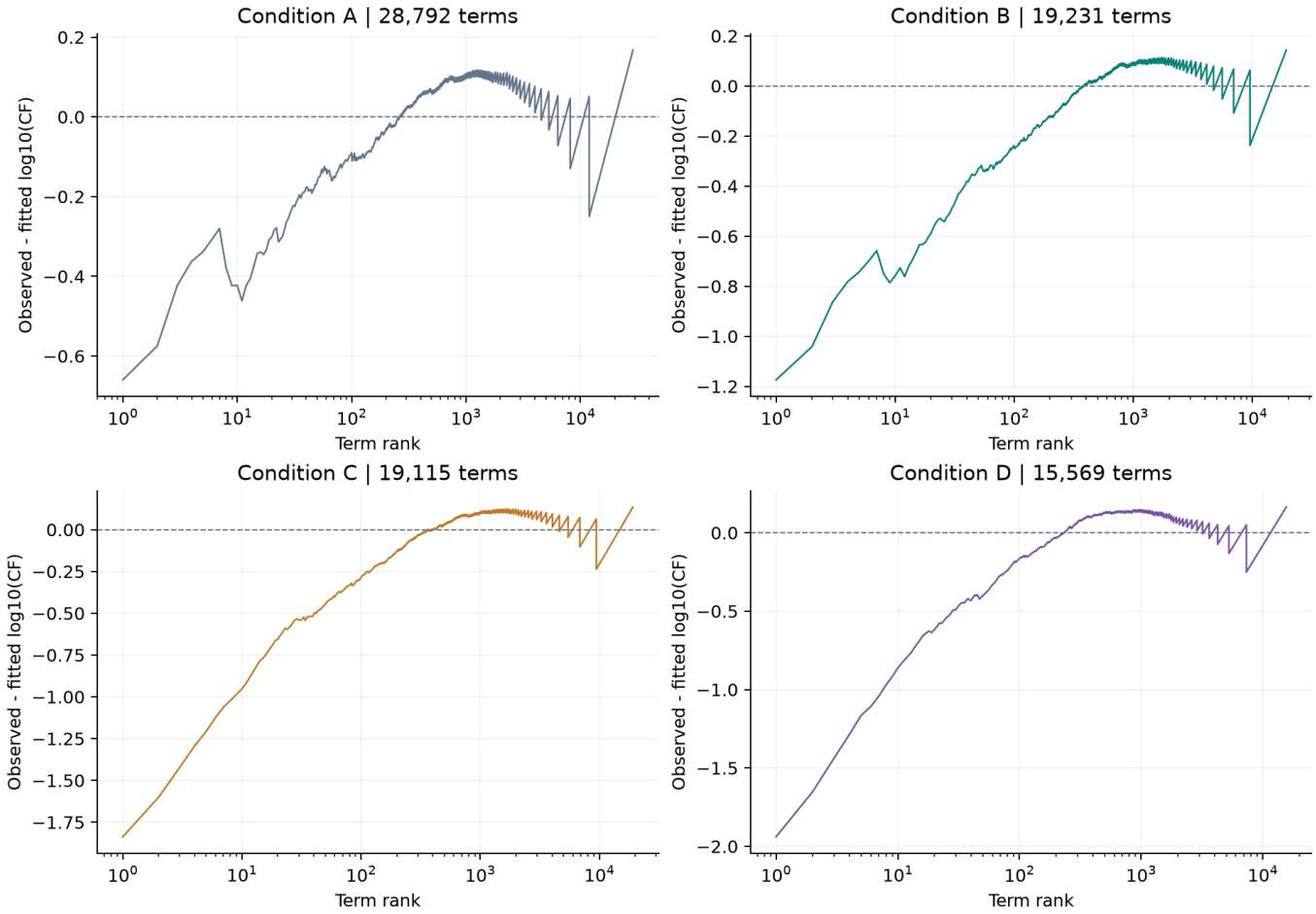
With log10(CF)=a+m*log10(rank), k=-m. Full-vocabulary estimates are A: 1.0915, B: 1.2508, C: 1.2095, D: 1.2663. Condition B has slope -1.2508, intercept 5.2149, and k=1.2508. The estimate depends on preprocessing and the fitted rank range. The 1/r reference is anchored to the most frequent term; it is not the fitted regression line.

Regional fit and residuals

Stage	Region	Ranks	k	R^2	RMSE
A	Head (0–10%)	1–2880	0.9568	0.9921	0.0368
A	Middle (10–80%)	2881–23034	1.1852	0.9236	0.0802
A	Tail (80–100%)	23035–28792	-0.0000	N/A	0.0000
B	Head (0–10%)	1–1924	0.9490	0.9900	0.0411
B	Middle (10–80%)	1925–15385	1.4790	0.9628	0.0685
B	Tail (80–100%)	15386–19231	-0.0000	N/A	0.0000
C	Head (0–10%)	1–1912	0.8447	0.9728	0.0608
C	Middle (10–80%)	1913–15292	1.4540	0.9614	0.0686
C	Tail (80–100%)	15293–19115	-0.0000	N/A	0.0000
D	Head (0–10%)	1–1557	0.9166	0.9579	0.0825
D	Middle (10–80%)	1558–12456	1.4525	0.9550	0.0742
D	Tail (80–100%)	12457–15569	-0.0000	N/A	0.0000

The rank-frequency curves have a high-frequency head and a long rare-term tail, with an approximately linear middle on log-log axes. For condition B, the full-vocabulary fit has R^2=0.9687 and log10 RMSE=0.0975. The exponent differs from the idealized value 1; the corpus is Zipf-like rather than an exact 1/r distribution. Among predefined regions with nonconstant frequencies, Head (0–10%) has the smallest log-space RMSE (0.0411, R^2=0.9900). The complete regional table and residuals show where the approximation changes. R^2 alone is insufficient: it measures a descriptive OLS fit after a logarithmic transformation. Ranked observations are dependent, many tail frequencies are tied, and no competing distribution or formal power-law goodness-of-fit test is evaluated. Constant tails have undefined R^2, not proof of an exact law.

Residuals from the full fit



Discrete low-frequency ties create a structured tail. A constant-frequency region has undefined R^2; a near-zero residual there is not evidence that the complete distribution is an exact power law.

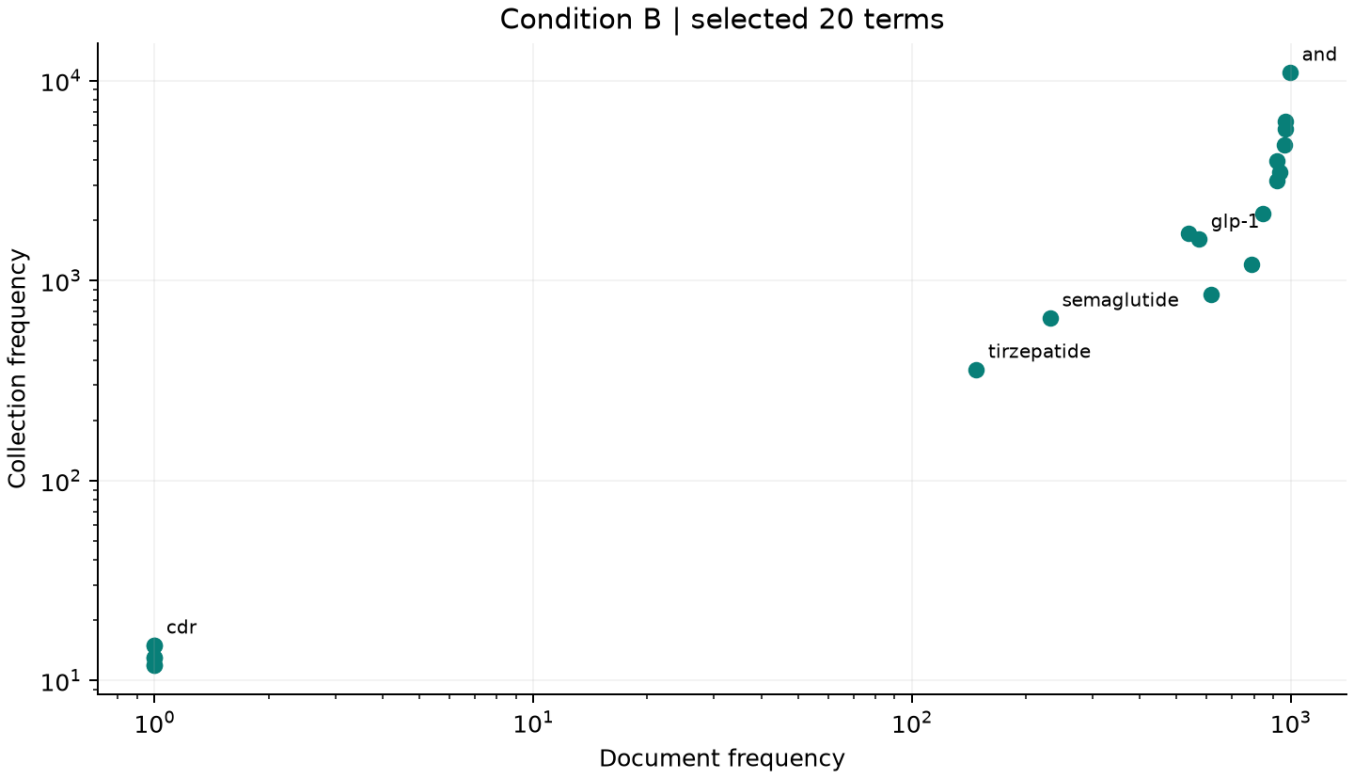
CF, DF and textbook IDF

Condition B; IDF = ln(N/DF), N = 1,000. Selection: top 10 CF, available biomedical terms, then high CF/DF terms with CF ≥ 5.

Term	CF	DF	CF/DF	IDF
and	10988	997	11.02	0.0030
the	6261	971	6.45	0.0294
of	5710	967	5.90	0.0336
in	4792	962	4.98	0.0387
with	3964	918	4.32	0.0856
to	3488	938	3.72	0.0640
a	3161	919	3.44	0.0845
for	2166	846	2.56	0.1672
were	1719	536	3.21	0.6236
glp-1	1606	573	2.80	0.5569
receptor	1200	787	1.52	0.2395
semaglutide	649	232	2.80	1.4610
tirzepatide	359	148	2.43	1.9105
agonists	848	615	1.38	0.4861
cdr	15	1	15.00	6.9078
eyes	15	1	15.00	6.9078
c3m	13	1	13.00	6.9078
std	13	1	13.00	6.9078
brca1	12	1	12.00	6.9078
ebmt	12	1	12.00	6.9078

CF counts occurrences; DF counts distinct documents. In condition B, 'and' occurs 10,988 times across 997 abstracts, with IDF=0.0030. In contrast, 'cdr' has CF=15, DF=1 and CF/DF=15.00: repetition is concentrated rather than widespread. A paper can repeat a specialized abbreviation many times, raising CF while contributing only one to DF. DF/N directly describes document coverage. Both measures matter: CF captures repetition and processing volume; DF determines postings-list coverage and collection-level discrimination. IDF=ln(N/DF) is smaller for terms occurring in many documents. CF alone cannot determine IDF, because equal CF can have different DF.

CF versus document coverage



Terms with similar CF may have different DF and therefore different IDF. Numeric tables and the interactive term-to-document view support inspection of actual repetition.

Information retrieval implications

Zipf's law matters because an information retrieval system processes a highly uneven vocabulary. In this collection, condition B contains 19,231 distinct terms and 243,843 occurrences. A small head contributes many occurrences, while numerous rare forms occupy the tail. This structure connects language statistics to indexing cost and term weighting. It is an empirical design guide rather than a rule that every biomedical collection must satisfy exactly.

Stopword removal changes the workload substantially: condition C retains 177,052 tokens, compared with 243,843 in B. Common grammatical words often occur in most documents, so their presence provides little evidence that one paper is more relevant than another. Removing them can shorten query processing and positional data. However, frequent words are not automatically useless: phrases, negation, and biomedical concepts may depend on them. The experiment demonstrates changed counts, not an improvement in retrieval effectiveness. Phrase search should therefore preserve the original text and use an explicitly documented matching policy.

For an inverted index, each unique term needs a dictionary entry and a postings list. High-DF terms have long document lists; high-CF terms may also have many stored positions. The long tail creates many short lists and dictionary entries, even when each term has few occurrences. Porter reduces vocabulary from 19,115 to 15,569 between C and D, suggesting a smaller dictionary, but merged lists may become longer. Actual storage also depends on document-ID gaps, position encoding and implementation overhead. Gap-based integer compression and compact dictionaries are relevant design options; this study does not claim measured compression savings.

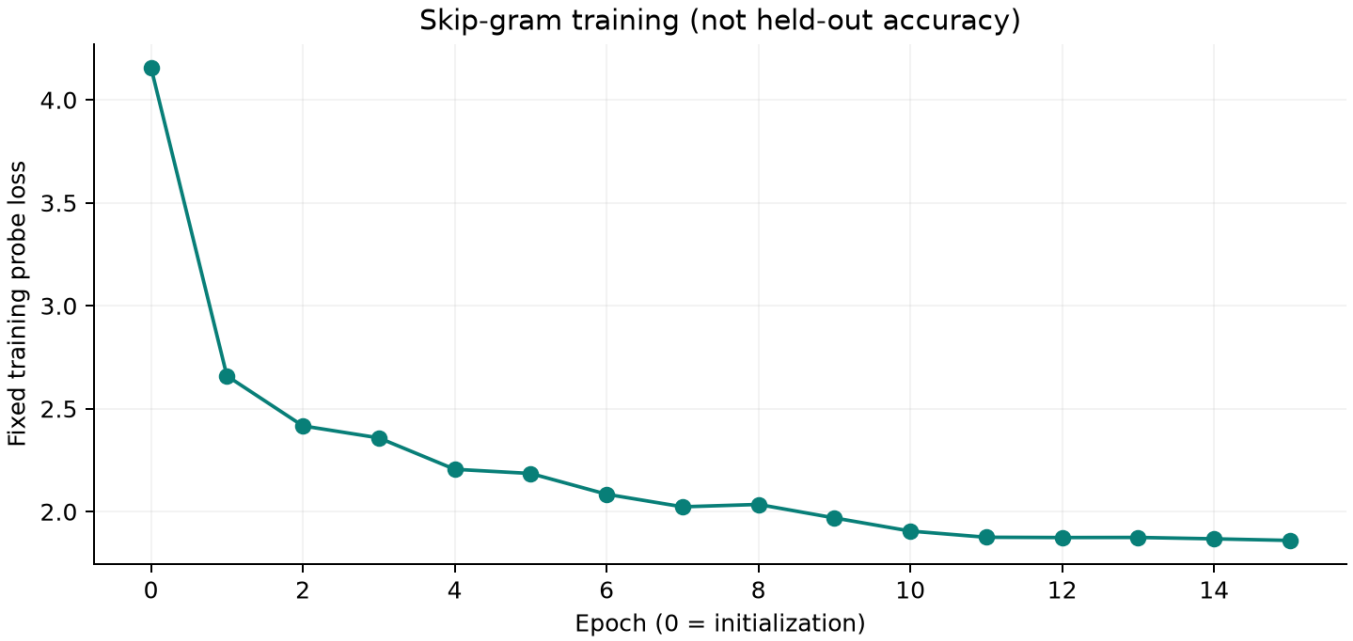
TF-IDF separates within-document repetition from collection-wide prevalence. Raw TF can reward a term repeated in a document, while IDF=ln(N/DF) discounts terms shared by many documents. A term found in every document has IDF zero under this formula. High CF alone does not imply low IDF: a specialized term can be repeated heavily in only a few abstracts. Comparing CF and DF reveals this distinction. Zipf's head helps explain why frequency alone is an inadequate ranking signal, but DF must be measured independently. Helix keeps BM25 ranking separate from this textbook-IDF analysis. Ultimately, frequency statistics justify concrete experiments on index size, query latency and relevance judgments; they cannot replace those evaluations.

Final challenge

The four conditions produce different exponents (A: 1.0915, B: 1.2508, C: 1.2095, D: 1.2663), yet all show strongly uneven term frequencies. That robust qualitative structure is useful even though no single exact law explains every rank. It motivates separate treatment of very common terms, compact storage for a large rare-term vocabulary, and weighting based on document coverage. These are engineering hypotheses supported by observed counts, not universal guarantees. This study measures vocabulary reduction and frequency structure; it does not measure compressed index bytes or retrieval relevance. A subsequent evaluation should compare latency, actual storage, and judged search quality before claiming an improvement. The useful lesson is to let the empirical curve guide design, then validate the design against the system's retrieval objective.

Word2Vec: Skip-gram

Condition C, 9,427 vocabulary words, 64 dimensions. Config: {"dimensions": 64, "window": 5, "min_count": 2, "negatives": 5, "epochs": 15, "learning_rate": 0.025, "sample": 0.001, "batch_size": 128, "seed": 42}



Probe loss uses fixed training examples and negative samples, including initialization. It is not held-out accuracy. Neighbors measure context similarity, not guaranteed biomedical synonymy.

Query	Five nearest words
glp-1	ras (0.796), substitution (0.698), ra (0.682), cornerstones (0.672), polypeptide/glucagon-like (0.670)
semaglutide	liraglutide (0.612), dulaglutide (0.607), once-weekly (0.569), 2·4 (0.563), avexitide (0.549)
tirzepatide	dulaglutide (0.644), cagrilintide-semaglutide (0.590), exenatide (0.574), liraglutide (0.572), 32.6% (0.570)

Limitations and reproducibility

The sample is one English biomedical topic selected in PubMed order, not a random cross-domain corpus. Tokenization retains internal compounds and numerical strings. Stemming can merge distinct meanings. There are no relevance labels or external embedding benchmark, so no retrieval-accuracy or semantic-quality claim is made.

Reproduce

Run .\.venv\python.exe scripts\run_project2.py. Source abstracts and hashes are in corpus_snapshot.json; run_metadata.json records settings, environment and file checksums. The exact stopword list is in analysis.json. Full frequencies and selected terms are CSV exports.

Sources

Course briefs: Homework-2-26-20260922.pdf; project-2-details-20260929.pdf. PubMed source records are identified by PMID in the corpus snapshot. Local implementation: src/zipf_experiment.py, src/porter.py and src/word2vec.py.

Appendix: Top 50 · condition A

Rank	Term	CF	DF	IDF
1	and	10965	997	0.0030
2	the	6256	971	0.0294
3	of	5709	967	0.0336
4	in	4788	962	0.0387
5	with	3961	918	0.0856
6	to	3487	938	0.0640
7	a	3142	918	0.0856
8	for	2162	846	0.1672
9	were	1716	535	0.6255
10	was	1534	553	0.5924
11	is	1264	638	0.4494
12	this	1256	768	0.2640
13	glp-1	1196	458	0.7809
14	as	1187	613	0.4894
15	receptor	1185	784	0.2433
16	or	1113	540	0.6162
17	by	1026	568	0.5656
18	that	992	527	0.6406
19	on	991	558	0.5834
20	weight	950	381	0.9650
21	patients	937	385	0.9545
22	metabolic	904	409	0.8940
23	are	794	510	0.6733
24	from	770	478	0.7381
25	=	750	184	1.6928
26	these	750	522	0.6501
27	risk	749	369	0.9970
28	glucagon-like	733	666	0.4065
29	clinical	727	433	0.8370
30	we	719	443	0.8142
31	associated	708	397	0.9238
32	peptide-1	680	624	0.4716
33	obesity	676	343	1.0700
34	evidence	665	395	0.9289

Rank	Term	CF	DF	IDF
35	agonists	659	519	0.6559
36	treatment	627	374	0.9835
37	2	624	450	0.7985
38	not	611	390	0.9416
39	may	597	378	0.9729
40	diabetes	594	436	0.8301
41	at	580	338	1.0847
42	an	552	403	0.9088
43	95%	539	160	1.8326
44	cardiovascular	530	228	1.4784
45	semaglutide	505	185	1.6874
46	type	503	391	0.9390
47	effects	495	320	1.1394
48	study	495	354	1.0385
49	including	484	376	0.9782
50	disease	482	274	1.2946

Appendix: Top 50 · condition B

Rank	Term	CF	DF	IDF
1	and	10988	997	0.0030
2	the	6261	971	0.0294
3	of	5710	967	0.0336
4	in	4792	962	0.0387
5	with	3964	918	0.0856
6	to	3488	938	0.0640
7	a	3161	919	0.0845
8	for	2166	846	0.1672
9	were	1719	536	0.6236
10	glp-1	1606	573	0.5569
11	was	1534	553	0.5924
12	is	1269	638	0.4494
13	this	1262	771	0.2601
14	or	1226	551	0.5960
15	receptor	1200	787	0.2395
16	as	1189	615	0.4861
17	patients	1104	425	0.8557
18	weight	1050	408	0.8965
19	by	1026	568	0.5656
20	that	999	530	0.6349
21	on	993	560	0.5798
22	obesity	977	459	0.7787
23	metabolic	951	424	0.8580
24	risk	911	415	0.8795
25	agonists	848	615	0.4861
26	diabetes	801	509	0.6753
27	are	795	511	0.6714
28	from	775	479	0.7361
29	treatment	761	418	0.8723
30	these	758	523	0.6482
31	clinical	745	443	0.8142
32	outcomes	742	400	0.9163
33	glucagon-like	741	673	0.3960
34	evidence	739	416	0.8771

Rank	Term	CF	DF	IDF
35	we	719	443	0.8142
36	associated	710	398	0.9213
37	peptide-1	697	639	0.4479
38	95%	689	201	1.6045
39	2	681	474	0.7465
40	disease	678	334	1.0966
41	effects	653	393	0.9339
42	semaglutide	649	232	1.4610
43	studies	644	333	1.0996
44	ci	640	179	1.7204
45	not	622	395	0.9289
46	may	599	380	0.9676
47	loss	587	292	1.2310
48	at	582	338	1.0847
49	p	576	165	1.8018
50	study	576	386	0.9519

Appendix: Top 50 · condition C

Rank	Term	CF	DF	IDF
1	glp-1	1606	573	0.5569
2	receptor	1200	787	0.2395
3	patients	1104	425	0.8557
4	weight	1050	408	0.8965
5	obesity	977	459	0.7787
6	metabolic	951	424	0.8580
7	risk	911	415	0.8795
8	agonists	848	615	0.4861
9	diabetes	801	509	0.6753
10	treatment	761	418	0.8723
11	clinical	745	443	0.8142
12	outcomes	742	400	0.9163
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18	2	681	474	0.7465
19	disease	678	334	1.0966
20	effects	653	393	0.9339
21	semaglutide	649	232	1.4610
22	studies	644	333	1.0996
23	ci	640	179	1.7204
24	may	599	380	0.9676
25	loss	587	292	1.2310
26	p	576	165	1.8018
27	study	576	386	0.9519
28	use	566	287	1.2483
29	cardiovascular	551	234	1.4524
30	type	521	401	0.9138
31	including	498	384	0.9571
32	therapy	486	255	1.3665
33	using	483	336	1.0906
34	review	445	337	1.0877

Rank	Term	CF	DF	IDF
35	trials	445	256	1.3626
36	adults	435	254	1.3704
37	among	426	261	1.3432
38	management	409	280	1.2730
39	insulin	397	202	1.5995
40	ra	397	132	2.0250
41	therapies	395	256	1.3626
42	body	379	225	1.4917
43	ras	379	167	1.7898
44	data	373	284	1.2588
45	significant	370	253	1.3744
46	included	359	249	1.3903
47	tirzepatide	359	148	1.9105
48	glp-1ra	353	122	2.1037
49	health	353	231	1.4653
50	across	347	255	1.3665

Appendix: Top 50 · condition D

Rank	Term	CF	DF	IDF
1	glp-1	1606	573	0.5569
2	us	1297	577	0.5499
3	patient	1271	485	0.7236
4	studi	1246	604	0.5042
5	receptor	1245	793	0.2319
6	agonist	1125	772	0.2588
7	weight	1081	423	0.8604
8	metabol	1073	463	0.7700
9	effect	1043	535	0.6255
10	obes	1039	468	0.7593
11	risk	1007	430	0.8440
12	associ	972	460	0.7765
13	includ	941	585	0.5361
14	outcom	937	451	0.7963
15	clinic	920	501	0.6911
16	diabet	915	529	0.6368
17	therapi	881	438	0.8255
18	treatment	848	451	0.7963
19	ra	776	204	1.5896
20	diseas	758	359	1.0244
21	evid	746	421	0.8651
22	glucagon-like	741	673	0.3960
23	peptide-1	697	639	0.4479
24	95%	689	201	1.6045
25	2	681	474	0.7465
26	ci	662	180	1.7148
27	improv	650	387	0.9493
28	semaglutid	649	232	1.4610
29	trial	634	328	1.1147
30	remain	608	464	0.7679
31	compar	606	314	1.1584
32	mai	599	380	0.9676
33	loss	591	293	1.2276
34	p	577	166	1.7958

Rank	Term	CF	DF	IDF
35	cardiovascular	551	234	1.4524
36	increas	543	358	1.0272
37	review	542	361	1.0189
38	type	530	407	0.8989
39	reduc	501	359	1.0244
40	control	483	320	1.1394
41	evalu	483	370	0.9943
42	manag	481	319	1.1426
43	adult	478	273	1.2983
44	reduct	458	267	1.3205
45	assess	454	325	1.1239
46	among	426	261	1.3432
47	insulin	403	203	1.5945
48	report	400	207	1.5750
49	signific	397	265	1.3280
50	potenti	390	308	1.1777